DSE-B-3

(Computational Mathematics)

Full Marks : 50

Answer ***question no.*** **1** and ***any four*** questions from the rest.

1. Answer ***any five*** questions : 2×5

(a) Define Hamiltonian circuit of a graph.

(b) What is a simple graph?

(c) What is a connected graph?

(d) Discuss the limitations of the Simpson's $1/3$ rd rule for solving definite integrals.

(e) What is the condition of convergence of the Gauss-Seidel method for solving a set of linear equations?

(f) What is the rank of a coefficient matrix in a system of linear equations?

(g) Is there any limitations of the Newton-Raphson's method? Explain your answer.

(h) If $f(x) = 4 \cos x - 6x$, find the relative percentage error in $f(x)$ for $x = 0$, if error in $x = 0.005$.

2. (a) What is Euler graph? Show that if G is an Euler connected graph then every vertex of G are of even degree.

(b) What is a tree? Show that a tree with n vertices has exactly $(n-1)$ edges. 5+5

3. (a) Define planar graph. Give an example.

(b) When are two graphs said to be isomorphic? Discuss with a suitable example.

(c) Show that the number of vertices of odd degree in a graph is always even. 2+3+5

4. (a) Solve the following system of linear equations by Gauss-Jordan Elimination method :

$$2x + 3y - 4z = 1$$

$$x - 2y + 3z = 2$$

$$-4x + y - 2z = 3$$

Correct up to two decimal places.

(b) Define absolute error, relative error and percentage error. 7+3

5. (a) Using the Bisection method, compute the real root of $x^3 - 3x^2 + 2x - 2 = 0$ correct up to two decimal places.

(b) Prove that $\Delta \cdot \nabla = \Delta - \nabla$, where $\Delta f(x) = f(x+h) - f(x)$ and $\nabla f(x) = f(x) - f(x-h)$. 6+4

6. Write down the composite expression for Simpson's 1/3 rd rule. Evaluate $\int_1^2 \frac{dx}{\sqrt{1+x^2}}$, by taking 8 intervals using this rule. Compute the error in this case. 2+6+2

7. (a) Given the following table, find $f(2.6)$ using Newton's backward interpolation polynomial technique.

x	0	1	2	3	4	5
y	−7	−9	−11	−7	9	43

(b) Discuss the geometrical interpolation of Newton-Raphson method with a diagram. 6+4

8. (a) Solve $x - 2\sin x - 3 = 0$ correct up to five significant figures by Newton-Raphson method.

(b) Solve the following system of linear equations by Gauss elimination method.

$$x - y + 2z = 4$$
$$-3x + 4y - z = 2$$
$$2x + 3y + 4z = 1$$

5+5

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